UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Computer generations and history

Artificial intelligence (A1) is the computer ability to exhibit human intelligent behaviors such as smart home applications, robots, intelligent tutoring systems and others.
a) In which generation is A1 introduced?
b) What is the programming language used for this technology?
c) What is the advantage of embedding A1 in our everyday life?
d) A1 can also posed as a threat to community. Discuss ONE (1) possible scenario when A1 can harm people.

Source: CSC429 (5).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Data representation and number systems

Convert 277410 to binary. Show your complete workflow.

Answer: 277410 = 1010110101102

- *1 mark for identifying the target base.*
- *1 mark for a correct repeated-division or place-value method.*
- *1 mark for the correct digit sequence.*
- *1 mark for the correct base subscript.*
- *1 mark for a complete, consistent workflow.*

QUESTION A4 (5 marks)

Topic: Parallel and multiprocessor architecture

Given the following Non Uniform Memory Access (NUMA) diagram.
2-1
1
'2
Figure 3: NUMA Architecture
a) State ONE (1) advantage of Symmetric Multi-Processor (SMP) that is applied in NUMA architecture.
b) Describe ONE (1) disadvantage of cluster architecture.
c) Draw a possible architecture for cluster architecture based on Figure 3.

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Source: CSC429 (6).pdf, page(s) 7

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Input/output and interrupts

Figure 1 illustrates the multiple interrupts that occur in a
Question 9 is based on Figure 1.
computer system.
User
Interrupt
Interrupt
Program
Handler W
Handler X
Interrupt
Handler Z
Interrupt
Handler Y
Figure 1: Multiple interrupts
a)
Define interrupt.
b)
Rank the interrupts based on their priority from the most important to the least important.
Rank the interrupts from the most important to the least important if disable interrupt is
c)
applied.

Source: CSC429 (15).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Computer architecture and CPU organisation

a) Compare system interconnection and CPU interconnection in terms of:
i) function
ii) physical appearance.
b) Describe the function of the THREE (3) different bus types in a computer system.

Source: CSC429 (16).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A7 (10 marks)

Topic: Memory hierarchy and cache

a) Why replacement algorithm for cache is needed?
b) Discuss all different types of available replacement.

Source: CSC429 (7).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Input/output and interrupts

a) Compare TWO (2) system bus timing approach with appropriate examples.
b) Discuss ONE (1) reason why an I/O module is needed.
c) List TWO (2) interfaces to external devices

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paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Computer architecture and CPU organisation

Processor contains a number of registers used for temporary storage of data. Describe FIVE (5) types of registers.

Source: CSC429 (2).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Data representation and number systems

Answer both parts for the same expression. Assume an 8-bit byte.
Expression: -24 - 63

a) Calculate -24 - 63 using sign and magnitude.

Answer: left -24 -> 1 0011000; right 63 -> 0 0111111; -24 - 63 = -87 -> sign-magnitude 1 1010111

- *1 mark for operand representations.*
- *1 mark for sign handling.*
- *1 mark for magnitude arithmetic.*
- *1 mark for the result sign bit.*
- *1 mark for the final byte.*

b) Calculate -24 - 63 using 1's and 2's complement.

Answer: left -24 -> 1's 11100111, 2's 11101000; right 63 -> 1's 00111111, 2's 00111111; -24 - 63 = -87 -> 1's 10101000, 2's 10101001

- *1 mark for the positive operand in binary.*
- *1 mark for the 1's complement of the negative operand.*
- *1 mark for adding 1 to form the 2's complement.*
- *1 mark for the correct 8-bit result.*
- *1 mark for converting the result back to decimal.*

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 5 and Table 1
z-A+c2/B*D-3A
Figure 5
Write an appropriate program using conect assembly language notations to compute the equation shown in Figure 5 using instructions for zero-address, one-address, and three-address. The available instructions are provided in Table 1. The registers labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register z.
Table 1: lt/achine instruction
Three-address
Zero-address
One-address
ADD (p,q,r)
LOAD p
PUSH p
SUB (p,q,r)
STOR p
POP p
MUL (p,q,r)
ADD p
ADD
DIV (p,q,r)
SUB p
SUB
MUL p
MUL
DIV p
DIV

Source: CSC429 (21).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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